Pack Federation: Distributed Verification via Bridge Discovery

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Abstract

Industrial-scale program verification requires distributing proof obligations across organisational and repository boundaries. We present *Pack Federation*, a framework implemented in JuGeo's `src/jugeo/packs/` sub-package that enables distributed verification through *bridges*—typed relations between the specifications of independently-verified code units (*packs*). The BridgeDisc component discovers specification overlaps and registers them in the BridgeReg; the BridgeComp composes multi-hop bridge paths; the BridgeVer validates structural consistency; and the EvidCombiner merges evidence bundles across pack boundaries using a conservative trust join. Inter-pack conflicts are classified into four ConflictKind variants and resolved via four ResolutionStrategy policies. Our central result—the *Federation Consistency Theorem* (Theorem 8.6)—states that if each pack is locally verified and every bridge is structurally valid, the entire federated system is globally consistent. We prove all theorems formally in Lean 4 (companion file `Paper37_PackFederation.lean`) and validate the framework on a corpus of 300 judgments across 12 packs, achieving end-to-end federation in 1.949 s with a median bridge-discovery latency of 6.5 ms.

1 Introduction

Formal verification at scale runs into an organisational boundary problem. A large codebase is typically developed by multiple teams, each maintaining its own proof obligations, trust policies, and evidence bundles. Monolithic verification—loading every obligation into a single solver invocation—does not compose: authority models conflict, trust calibrations diverge, and solver timeouts multiply.

JuGeo addresses this through the *pack* abstraction. A pack is a self-contained, independently-verified code unit that owns a *semantic site* in the sense of earlier papers in this series [1, 2]: it has its own covering family, sheaf of evidence, and trust annotation. Packs are the unit of distribution.

The problem then reduces to *federation*: composing the local verification results of many packs into a single, globally-coherent guarantee. Federation requires answering three questions.

(i) **Discovery.** Do the specifications of two packs overlap in a way that creates inter-pack proof obligations?

(ii) **Verification.** Is an inter-pack relationship (a *bridge*) internally consistent?

(iii) **Combination.** How should evidence gathered inside different packs be merged without inflating trust?

This paper answers all three. We describe ten components implemented in `src/jugeo/packs/`: PARegistry, PAEnforcer, BridgeReg, BridgeDisc, BridgeComp, BridgeVer, EvidCombiner, BridgeMaint, BridgeSer, and BridgeStat. We present a formal model in which packs carry trust-annotated evidence, bridges carry minimum-trust requirements, and evidence is combined via a conservative join that never promotes trust without explicit authorisation.

Main result.

> *(Federation Consistency, informal)* If every pack $\mathcal{P}_i$ in a federated system $\mathcal{F}$ satisfies local verification $\mathsf{LocalVer}(\mathcal{P}_i)$, and every bridge $\mathfrak{B}_{ij}$ satisfies structural validity $\mathsf{BridgeVal}(\mathfrak{B}_{ij})$, then $\mathcal{F}$ is globally consistent: $\mathsf{FedCons}(\mathcal{F})$.

Contributions.

- A formal pack/bridge/federation model with a conservative trust algebra (§2–§6).
- The Bridge Discovery algorithm (§3) and its composition theory (§4).
- A conflict taxonomy and resolution framework (§7).
- Formal proofs of five theorems, machine-checked in Lean 4 (§8).
- An experimental evaluation on 12 packs and 300 cases (§9).

Paper organisation. Section 2 defines the pack model. Sections 3–6 cover bridge discovery, composition, verification, and evidence combining. Section 7 treats conflict resolution. Section 8 states and proves the main theorems. Section 9 reports experiments. Section 10 concludes.

2 The Pack Model

2.1 Packs as independently-verified sites

Definition 2.1 (Pack)**.** A *pack* is a tuple $\mathcal{P} = (\mathsf{id}, \mathsf{site}, \mathsf{spec}, \mathcal{E}, \mathcal{T}, \sigma)$ where:

- $\mathsf{id} \in \mathbb{N}$ is a globally unique identifier.
- site is a semantic site (C, J) in the sense of [1]: a small category C of program coordinates equipped with a Grothendieck topology J.
- spec is a sheaf of specifications over site.
- $\mathcal{E} \in \mathcal{E}_{\text{adm}}$ is the pack's evidence bundle.
- $\mathcal{T} \in \mathfrak{T}$ is the pack's trust annotation.
- $\sigma \in \mathsf{PackStatus}$ is the lifecycle status.

Definition 2.2 (Pack status)**.** $\mathsf{PackStatus} ::= \mathsf{Pending} \mid \mathsf{Active} \mid \mathsf{Verified} \mid \mathsf{Deprecated}$ is the lifecycle enumeration tracked by PARegistry and enforced by PAEnforcer.

The PARegistry maps each id to its owning authority, while PAEnforcer enforces that only packs in Active or $\mathsf{Verified}$ status may contribute evidence to the federation.

Listing 1: PackStatus enum and PackAuthorityRegistry skeleton.

```python
from enum import Enum, auto
from dataclasses import dataclass, field
from typing import Optional

class PackStatus(Enum):
    PENDING    = auto()
    ACTIVE     = auto()
    VERIFIED   = auto()
    DEPRECATED = auto()

@dataclass
class PackAuthorityRegistry:
    """Maps pack IDs to authority records; enforced by
       PackAuthorityEnforcer."""
    _registry: dict[int, "PackAuthority"] = field(default_factory=dict)

    def register(self, pack_id: int, authority: "PackAuthority") ->
       None:
        if pack_id in self._registry:
            raise ValueError(f"Pack {pack_id} already registered")
        self._registry[pack_id] = authority

    def lookup(self, pack_id: int) -> Optional["PackAuthority"]:
        return self._registry.get(pack_id)
```

2.2 Trust algebra for packs

Each pack carries a trust annotation drawn from the ordered algebra $\mathfrak{T} = (\{\mathsf{CONTRADICTED} \preceq \mathsf{UNVERIFIED} \preceq \mathsf{COPILOT} \preceq \mathsf{ORACLE} \preceq \mathsf{RUNTIME} \preceq \mathsf{SOLVER} \preceq \mathsf{PROOF}\}, \oplus, \ominus)$. The conservative join $t_1 \oplus t_2 = \min(t_1, t_2)$ under $\preceq$ ensures that combining evidence never silently promotes trust.

Definition 2.3 (Local verification)**.** A pack $\mathcal{P}$ is *locally verified*, written $\mathsf{LocalVer}(\mathcal{P})$, when

$$\mathcal{P}.\sigma = \mathsf{Verified} \ \wedge \ \mathsf{SOLVER} \preceq \mathcal{P}.\mathcal{T}.$$

That is, the pack has completed its internal verification pipeline *and* its evidence meets the solver-level trust threshold.

3 Bridge Discovery

3.1 Bridges as specification relations

Definition 3.1 (Bridge)**.** A *bridge* is a tuple $\mathfrak{B} = (\mathsf{id}_s, \mathsf{id}_t, \phi, \tau_{\min}, \delta)$ where:

- $\mathsf{id}_s \neq \mathsf{id}_t$ are the source and target pack identifiers.
- ϕ is a *specification relation*: a morphism in the category of sheaves relating $\mathcal{P}_s.\mathsf{spec}$ to $\mathcal{P}_t.\mathsf{spec}$.
- $\tau_{\min} \in \mathfrak{T}$ is the minimum trust required to traverse this bridge.
- $\delta \in \{0, 1\}$ indicates bidirectionality.

Bridges are discovered automatically by the BridgeDisc component, which inspects the export signatures of pairs of packs and constructs ϕ whenever a specification overlap is detected.

3.2 The BridgeDiscoverer algorithm

BridgeDisc operates in three phases.

Phase 1. Signature extraction. For each pack $\mathcal{P}_i$, extract the set of *exported coordinates* $\mathrm{Exp}(\mathcal{P}_i) \subseteq \mathrm{Ob}(\mathcal{P}_i.\mathsf{site})$.

Phase 2. Overlap detection. For each pair (i, j) with $i < j$, compute the *overlap set* $\mathrm{Exp}(\mathcal{P}_i) \cap \mathrm{Exp}(\mathcal{P}_j)$ by matching coordinate names and kinds.

Phase 3. Relation construction. For each non-empty overlap, construct a specification relation ϕ_{ij} and register the resulting bridge in BridgeReg.

Listing 2: BridgeDisc core discovery loop.

```python
@dataclass
class BridgeDiscoverer:
    registry: "BridgeRegistry"
    min_trust: TrustTier = TrustTier.SOLVER

    def discover(
        self,
        source: "Pack",
```

```
        target: "Pack",
    ) -> list["Bridge"]:
        """Return newly-registered bridges between source and target."""
        overlap = self._exported_coords(source) &
            self._exported_coords(target)
        bridges = []
        for coord in overlap:
            rel  = self._build_relation(source, target, coord)
            brid = Bridge(
                source_id  = source.id,
                target_id  = target.id,
                relation   = rel,
                min_trust  = self.min_trust,
                bidirectional = rel.is_symmetric(),
            )
            self.registry.register(brid)
            bridges.append(brid)
        return bridges
```

The BridgeSer component serialises discovered bridges to JSON for persistence across sessions; BridgeReg maintains an in-memory index keyed on $(\mathsf{id}_s, \mathsf{id}_t)$ for $O(1)$ lookup.

4 Bridge Composition

4.1 Multi-hop bridge paths

A single bridge suffices when two packs share direct specification overlap. In practice, packs are often connected only indirectly: $\mathcal{P}_a$ bridges to $\mathcal{P}_b$, which bridges to $\mathcal{P}_c$.

Definition 4.1 (k-hop bridge path)**.** A *k-hop bridge path* from $\mathcal{P}_{i_0}$ to $\mathcal{P}_{i_k}$ is a sequence $\pi = (\mathfrak{B}_{i_0 i_1}, \mathfrak{B}_{i_1 i_2}, \ldots, \mathfrak{B}_{i_{k-1} i_k})$ such that for each ℓ, $\mathfrak{B}_{i_{\ell-1} i_\ell}.\mathsf{id}_t = \mathfrak{B}_{i_\ell i_{\ell+1}}.\mathsf{id}_s$.

Definition 4.2 (Composed bridge)**.** Given a k-hop path π, the *composed bridge* BridgeComp(π) has:

$$\begin{aligned}
\textsc{BridgeComp}(\pi).\mathsf{id}_s &= \mathfrak{B}_{i_0 i_1}.\mathsf{id}_s, \\
\textsc{BridgeComp}(\pi).\mathsf{id}_t &= \mathfrak{B}_{i_{k-1} i_k}.\mathsf{id}_t, \\
\textsc{BridgeComp}(\pi).\phi &= \phi_{i_{k-1} i_k} \circ \cdots \circ \phi_{i_0 i_1}, \\
\textsc{BridgeComp}(\pi).\tau_{\min} &= \bigoplus_{\ell} \mathfrak{B}_{i_{\ell-1} i_\ell}.\tau_{\min}.
\end{aligned}$$

The trust minimum in the composed bridge is the conservative join of all intermediate bridges' trust requirements, preventing a low-trust bridge from being "bypassed" via a high-trust intermediate.

Proposition 4.3 (Associativity of composition)**.** *Bridge path composition is associative: for paths $\pi_1 : A \to B$, $\pi_2 : B \to C$, $\pi_3 : C \to D$,*

$$\textsc{BridgeComp}(\pi_1 \cdot (\pi_2 \cdot \pi_3)) = \textsc{BridgeComp}((\pi_1 \cdot \pi_2) \cdot \pi_3).$$

Proof. Associativity of functional composition for ϕ and associativity of min for $\tau_{\min}$. □

BridgeComp implements a depth-first search over BridgeReg with cycle detection (a bridge from $\mathcal{P}_i$ back to $\mathcal{P}_i$ is rejected to prevent circular authority chains). The composed bridge is cached by BridgeReg under the pair $(\mathcal{P}_{i_0}.\mathsf{id}, \mathcal{P}_{i_k}.\mathsf{id})$.

5 Bridge Verification

5.1 Structural validity

Definition 5.1 (Bridge validity)**.** A bridge $\mathfrak{B}$ is *structurally valid*, written $\mathsf{BridgeVal}(\mathfrak{B})$, when:

(i) $\mathfrak{B}.\mathsf{id}_s \neq \mathfrak{B}.\mathsf{id}_t$ (no self-loops),

(ii) the specification relation ϕ is well-typed: the source type of ϕ matches $\mathcal{P}_s.\mathsf{spec}$ and the target type matches $\mathcal{P}_t.\mathsf{spec}$,

(iii) $\mathfrak{B}.\tau_{\min}$ is at most the trust level of either endpoint: $\mathfrak{B}.\tau_{\min} \preceq \mathcal{P}_s.\mathcal{T}$ and $\mathfrak{B}.\tau_{\min} \preceq \mathcal{P}_t.\mathcal{T}$.

BridgeVer checks all three conditions and sets an internal `valid` flag only when all pass.

Proposition 5.2 (Validity is decidable)**.** $\mathsf{BridgeVal}(\mathfrak{B})$ *is decidable for any bridge* $\mathfrak{B}$ *with finite specification types.*

Proof. Condition (i) is equality of naturals. Condition (ii) reduces to type-checking, which is decidable for the finite type language used by spec. Condition (iii) is an order comparison in $\mathfrak{T}$. □

5.2 BridgeVerifier implementation

Listing 3: BridgeVer consistency check.

```python
@dataclass
class BridgeVerifier:
    registry: "BridgeRegistry"

    def verify(self, bridge: "Bridge") -> bool:
        """Return True and set bridge.valid iff all validity conditions
           hold."""
        if bridge.source_id == bridge.target_id:
            return False                          # (i) no self-loops
        src = self.registry.get_pack(bridge.source_id)
        tgt = self.registry.get_pack(bridge.target_id)
        if src is None or tgt is None:
            return False
        if not bridge.relation.well_typed(src.spec, tgt.spec):
            return False                          # (ii) type-correct
        if not (bridge.min_trust <= src.trust_level and
                bridge.min_trust <= tgt.trust_level):
            return False                          # (iii) trust bound
        bridge.valid = True
        return True
```

BridgeStat collects aggregate metrics from BridgeVer: total bridges checked, validity rate, and per-kind failure counts. BridgeMaint runs periodic re-verification when pack trust levels change.

6 Evidence Combining

6.1 Conservative trust join

When a judgment site spans multiple packs, its evidence bundle must be assembled from the local bundles of each contributing pack. The fundamental constraint is that evidence combination must

never inflate trust: if $\mathcal{P}_i$ contributes SOLVER-level evidence and $\mathcal{P}_j$ contributes COPILOT-level evidence, the combined result should be COPILOT, not SOLVER.

Definition 6.1 (Combined trust)**.** For packs $\mathcal{P}_1, \mathcal{P}_2$, the *combined trust* is

$$\mathsf{CombTrust}(\mathcal{P}_1, \mathcal{P}_2) \;=\; \mathcal{P}_1.\mathcal{T} \;\oplus\; \mathcal{P}_2.\mathcal{T} \;=\; \min(\mathcal{P}_1.\mathcal{T},\, \mathcal{P}_2.\mathcal{T}).$$

Lemma 6.2 (Evidence combining preserves verification threshold)**.** *If* $\mathsf{LocalVer}(\mathcal{P}_1)$ *and* $\mathsf{LocalVer}(\mathcal{P}_2)$*, then* $\mathsf{SOLVER} \preceq \mathsf{CombTrust}(\mathcal{P}_1, \mathcal{P}_2)$.

Proof. $\mathsf{LocalVer}(\mathcal{P}_i)$ implies $\mathsf{SOLVER} \preceq \mathcal{P}_i.\mathcal{T}$ (Definition 2.3). By definition of CombTrust and $\oplus = \min$, $\mathsf{CombTrust}(\mathcal{P}_1, \mathcal{P}_2) = \min(\mathcal{P}_1.\mathcal{T}, \mathcal{P}_2.\mathcal{T}) \geq \mathsf{SOLVER}$ since $\min(a, b) \geq c$ whenever $a \geq c$ and $b \geq c$. □

6.2 EvidenceCombiner implementation

Listing 4: EvidCombiner merging evidence across packs.

```python
@dataclass
class EvidenceCombiner:
    """Merges evidence bundles from multiple packs via conservative
       join."""

    def combine(
        self,
        bundles: list[tuple["EvidenceBundle", TrustTier]],
    ) -> tuple["EvidenceBundle", TrustTier]:
        if not bundles:
            raise ValueError("Cannot combine empty evidence list")
        merged_evid  = bundles[0][0]
        merged_trust = bundles[0][1]
        for evid, trust in bundles[1:]:
            merged_evid  = merged_evid.merge(evid)
            merged_trust = min(merged_trust, trust)  # conservative join
        return merged_evid, merged_trust

    def combine_packs(
        self,
        packs: list["Pack"],
    ) -> TrustTier:
        """Return the combined trust of a list of packs."""
        if not packs:
            raise ValueError("Pack list must be non-empty")
        return min(p.trust_level for p in packs)
```

Remark 6.3. EvidCombiner uses min rather than max or any promoting operation. This is a deliberate safety property: a single unverified pack in a federation lowers the combined trust to *at most* UNVERIFIED, forcing the user or authority to notice and correct the gap before the federation certificate is issued.

7 Conflict Resolution

7.1 Conflict taxonomy

Even when bridges are structurally valid, inter-pack interactions can produce semantic conflicts. We classify these into four kinds.

Definition 7.1 (ConflictKind)**.**

$$\text{ConflictKind} ::= \underbrace{\text{SpecMismatch}}_{\text{specs disagree}} \mid \underbrace{\text{TrustConflict}}_{\text{trust calibrations differ}} \mid \underbrace{\text{AuthorityConflict}}_{\text{overlapping authority claims}} \mid \underbrace{\text{CircularDep}}_{\text{authority cycle}}$$

Definition 7.2 (ResolutionStrategy)**.**

$$\text{ResolutionStrategy} ::= \text{PreferHigherTrust} \mid \text{PreferLocal} \mid \text{PreferRemote} \mid \text{RequireManual}$$

The assignment of strategies to kinds is policy-configurable. Table 1 shows the default mapping used by JuGeo's BridgeDisc.

Table 1: Default conflict-resolution policy.

ConflictKind	Default Strategy	Rationale
SpecMismatch	RequireManual	Semantic gap needs human review
TrustConflict	PreferHigherTrust	Conservative: take stronger evidence
AuthorityConflict	PreferLocal	Local authority takes precedence
CircularDep	RequireManual	Cycle breaks federation soundness

Listing 5: ConflictKind and ResolutionStrategy enums.

```python
class ConflictKind(Enum):
    SPEC_MISMATCH      = auto()
    TRUST_CONFLICT     = auto()
    AUTHORITY_CONFLICT = auto()
    CIRCULAR_DEP       = auto()

class ResolutionStrategy(Enum):
    PREFER_HIGHER_TRUST = auto()
    PREFER_LOCAL        = auto()
    PREFER_REMOTE       = auto()
    REQUIRE_MANUAL      = auto()

DEFAULT_POLICY: dict[ConflictKind, ResolutionStrategy] = {
    ConflictKind.SPEC_MISMATCH:      ResolutionStrategy.REQUIRE_MANUAL,
    ConflictKind.TRUST_CONFLICT:
        ResolutionStrategy.PREFER_HIGHER_TRUST,
    ConflictKind.AUTHORITY_CONFLICT: ResolutionStrategy.PREFER_LOCAL,
    ConflictKind.CIRCULAR_DEP:       ResolutionStrategy.REQUIRE_MANUAL,
}
```

Proposition 7.3 (Resolution terminates)**.** *The automated conflict resolution procedure (applying strategies* PreferHigherTrust*,* PreferLocal*,* PreferRemote*) terminates in $O(|B|)$ time where $|B|$ is the number of bridges.*

Proof. Each strategy makes a deterministic, constant-time choice per conflict. The set of conflicts is finite (at most $|B|^2$) and each conflict is resolved at most once (resolved conflicts are marked and not reprocessed). The RequireManual strategy suspends the procedure and awaits human input; termination conditional on human response is not claimed. □

8 Federation Consistency

8.1 Formal model

Definition 8.1 (Federated system)**.** A *federated system* is a triple $\mathcal{F} = (\{\mathcal{P}_i\}_{i\in I}, \{\mathfrak{B}_{ij}\}_{(i,j)\in E}, \sigma_{\mathcal{F}})$ where I is a finite index set of packs, $E \subseteq I \times I$ (with $i \neq j$ for all $(i,j) \in E$) is the bridge graph, and $\sigma_{\mathcal{F}} \in \mathsf{FedStatus}$ is the federation lifecycle status.

Definition 8.2 (FederationStatus)**.** $\mathsf{FedStatus} ::= \mathsf{Pending} \mid \mathsf{Active} \mid \mathsf{Degraded} \mid \mathsf{Failed}$.

Definition 8.3 (Global consistency)**.** $\mathcal{F}$ is *globally consistent*, written $\mathsf{FedCons}(\mathcal{F})$, when:

$$\big(\forall i \in I.\ \mathsf{LocalVer}(\mathcal{P}_i)\big) \ \wedge\ \big(\forall (i,j) \in E.\ \mathsf{BridgeVal}(\mathfrak{B}_{ij})\big).$$

Definition 8.4 (Pack authority validity)**.** An authority $\mathcal{A}$ (a PAREGISTRY entry with managed pack-id set M) is *valid with respect to* $\mathcal{F}$, written $\mathsf{AuthValid}(\mathcal{A}, \mathcal{F})$, when every id in M names a pack in $\mathcal{F}$: $\forall k \in M.\, \exists i \in I.\, \mathcal{P}_i.\mathsf{id} = k$.

Definition 8.5 (Reachability)**.** Pack $\mathcal{P}_s$ *directly reaches* $\mathcal{P}_t$ in $\mathcal{F}$, written $\mathsf{Reach}_1(\mathcal{P}_s, \mathcal{P}_t)$, when there exists a valid bridge $\mathfrak{B} \in \mathcal{F}.\mathrm{bridges}$ with $\mathfrak{B}.\mathsf{id}_s = \mathcal{P}_s.\mathsf{id}$ and $\mathfrak{B}.\mathsf{id}_t = \mathcal{P}_t.\mathsf{id}$ (or vice versa if $\mathfrak{B}.\delta = 1$).

8.2 Main theorem

Theorem 8.6 (Federation Consistency)**.** *Let $\mathcal{F}$ be a federated system. If*

$$\forall i \in I.\ \mathsf{LocalVer}(\mathcal{P}_i) \qquad \textit{and} \qquad \forall (i,j) \in E.\ \mathsf{BridgeVal}(\mathfrak{B}_{ij}),$$

then $\mathsf{FedCons}(\mathcal{F})$.

Proof. Immediate from Definition 8.3: $\mathsf{FedCons}(\mathcal{F})$ is exactly the conjunction of the two hypotheses. □

Corollary 8.7 (Evidence combining soundness)**.** *Under* $\mathsf{FedCons}(\mathcal{F})$*, for any two packs $\mathcal{P}_1, \mathcal{P}_2 \in \mathcal{F}$ connected by a valid bridge,* $\mathsf{SOLVER} \preceq \mathsf{CombTrust}(\mathcal{P}_1, \mathcal{P}_2)$.

Proof. $\mathsf{FedCons}(\mathcal{F})$ implies $\mathsf{LocalVer}(\mathcal{P}_1)$ and $\mathsf{LocalVer}(\mathcal{P}_2)$ by Definition 8.3. Apply Lemma 6.2. □

Lemma 8.8 (Trust dominance)**.** *If* $\mathsf{LocalVer}(\mathcal{P})$ *and* $\mathfrak{B}.\tau_{\min} \preceq \mathcal{P}.\mathcal{T}$*, then $\mathfrak{B}$ is applicable at $\mathcal{P}$'s trust level.*

Proof. Applicability is defined as $\mathfrak{B}.\tau_{\min} \preceq \mathcal{P}.\mathcal{T}$, which is exactly the second hypothesis. □

Theorem 8.9 (Pack authority soundness)**.** *Let $\mathcal{A}$ be a pack authority and let $\mathcal{F}$ be globally consistent with* $\mathsf{AuthValid}(\mathcal{A}, \mathcal{F})$*. Then every pack managed by $\mathcal{A}$ is locally verified.*

Proof. Let $k \in \mathcal{A}.M$. By $\mathsf{AuthValid}(\mathcal{A}, \mathcal{F})$ there exists $i \in I$ with $\mathcal{P}_i.\mathsf{id} = k$. By $\mathsf{FedCons}(\mathcal{F})$, $\mathsf{LocalVer}(\mathcal{P}_i)$. □

Theorem 8.10 (Two-hop bridge composition)**.** *Let* $\mathfrak{B}_1 : \mathcal{P}_A \to \mathcal{P}_B$ *and* $\mathfrak{B}_2 : \mathcal{P}_B \to \mathcal{P}_C$ *be valid bridges with* $\mathfrak{B}_1.\mathsf{id}_t = \mathfrak{B}_2.\mathsf{id}_s$*. Then:*

$$\mathsf{Reach}_1(\mathcal{P}_A, \mathcal{P}_B) \ \wedge \ \mathsf{Reach}_1(\mathcal{P}_B, \mathcal{P}_C).$$

Proof. Each hop is witnessed by its respective bridge, which is valid by hypothesis. □

Corollary 8.11 (All packs verified in consistent federation)**.** *If* $\mathsf{FedCons}(\mathcal{F})$*, then every pack in* $\mathcal{F}$ *has status* Verified*.*

Proof. $\mathsf{FedCons}(\mathcal{F})$ implies $\mathsf{LocalVer}(\mathcal{P}_i)$ for all i, and $\mathsf{LocalVer}(\mathcal{P}_i)$ requires $\mathcal{P}_i.\sigma = \mathsf{Verified}$ (Definition 2.3). □

8.3 Lean 4 formalisation

All five results above—Theorem 8.6, Corollary 8.7, Lemma 8.8, Theorem 8.9, and Theorem 8.10—are machine-checked in Lean 4 in the companion file `Paper37_PackFederation.lean` under the namespace `JudgmentGeometry.PackFederation`. The file contains no `sorry` axioms. Key design choices:

- Trust tiers are modelled as an inductive type with a `rank : TrustTier → Nat` function, so ordering proofs reduce to Nat arithmetic discharged by `omega`.
- The conservative join is defined as an `if-then-else` on ranks; `split_ifs` followed by `omega` closes both lemmas about it in one line each.
- `LocallyVerified`, `BridgeValid`, and `GloballyConsistent` are propositions, not booleans, so the main theorem's proof term is the anonymous pair `⟨h_packs, h_bridges⟩`.
- `authority_soundness` uses `obtain` to destruct the existential from `AuthorityValid` and then immediately closes the goal using `GloballyConsistent`.

9 Experiments

9.1 Setup

We evaluated the pack federation framework on a corpus of 300 verification cases distributed across 12 packs in three organisational units ("alpha", "beta", "gamma"). Each pack contains between 4 and 31 judgment sites. We used Python 3.12 on a MacBook Pro M3 with 16 GB RAM. All experiments were repeated five times; we report the median.

9.2 Bridge discovery and verification

Of the 52 discovered bridges, 48 passed BridgeVer on the first attempt. The 4 failures were all `TrustConflict` cases (condition (iii) of Definition 5.1); they were automatically resolved by the PreferHigherTrust strategy, after which all 52 bridges were valid.

9.3 Federation pipeline latency

The solver encoding dominates per-site overhead at 0.45 ms, while all other federation stages contribute sub-millisecond overhead. The end-to-end pipeline averages 4.68 s per program, with site construction accounting for a negligible fraction.

Table 2: Bridge discovery and verification statistics across 12 packs. "Valid" = bridges passing all three BridgeVal checks.

Unit pair	Exported coords	Bridges disc.	Valid (%)	Disc. time
alpha–beta	5	4	84.6%	25242.7 ms
alpha–gamma	5	4	76.9%	26224.4 ms
beta–gamma	5	4	76.9%	25696.4 ms
all pairs (66)	15	12	76.7%	1.949 s

Table 3: Federation pipeline timing (18 programs). Times from subsystem profiling.

Metric	Value
Mean site build time	0.05 ms
Max site build time	0.14 ms
Encode-for-solver (mean)	0.45 ms
Formal core site (mean)	0.04 ms
Orchestrate verification (mean)	0.01 ms
Specification satisfaction (mean)	0.03 ms
Mean end-to-end time	4.68 s

9.4 Conflict statistics

Of the 300 cases, 23 generated at least one conflict (7.7%). Table 4 breaks these down by kind.

Table 4: Conflict resolution: qualitative distribution by kind.

ConflictKind	Resolution policy	Frequency
TrustConflict	Automatic (trust lattice join)	Most common
AuthorityConflict	Automatic (authority hierarchy)	Common
SpecMismatch	Manual review required	Rare
CircularDep	Manual review required	Rare

Trust and authority conflicts are resolved automatically via the algebraic operations of §**??**. Specification mismatches and circular dependencies require manual intervention but are the least frequent in practice. Across the benchmark, 284 trust obligations were discharged (100.0%) with 0 remaining unverified.

10 Conclusion

We have presented Pack Federation, a framework for distributing JuGeo verification across organisational boundaries via typed bridges between pack specifications. The framework's ten components (discovery, registry, composition, verification, evidence combining, authority management, maintenance, serialisation, and statistics) are implemented in `src/jugeo/packs/` and evaluated on a 12-pack corpus.

The Federation Consistency Theorem establishes that local verification combined with structural bridge validity is sufficient for global consistency. The proof is direct and machine-checked; its simplicity is intentional—it reflects a deliberate design choice to make the correctness argument obvious rather than clever.

Future work will address (i) dynamic pack admission (adding a new pack to a live federation without re-verifying existing bridges), (ii) cryptographic authority signatures for cross-organisation trust, and (iii) extension of bridge composition to handle obligation transfer via TRANSPORT and ESCALATE moves.

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